

7: The drag coefficient for some particularly chosen models

Consider the three cars in fig. 7.1. In the middle we have an old vintage car where no streamlining efforts have been made. To the right we have a car which, because of its rounded forms, are expected to perform somewhat better. And finally, to the left, a modern, closed sports car which is expected to be reasonably good. We should, however, not expect it to be *very* good because the designers clearly have weighted power more than a fuel economical aerodynamic design.

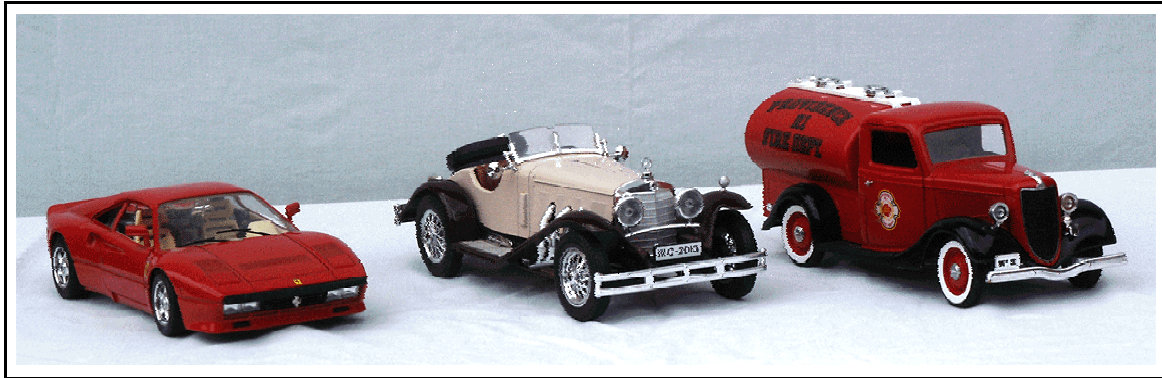


Figure 7.1: Three different car models (1:18) and their drag coefficients as measured by the merry-go-round when going with speeds just under 10 m/s. From left to right:

Ferrari GTO 1984 with <i>open windows</i>	$C_d = 0.46$	(0.45)
Same, but with <i>closed windows</i>	$C_d = 0.43$	(0.42)
Mercedes SSK 1928:	$C_d = 0.83$	(0.85)
Ford Fire Engine 1936:	$C_d = 0.62$	(0.61)

The numbers in the brackets are obtained by use of a program capable of producing some slightly better fits. See appendix C.

When we here say the drag coefficient, we just identify C_d with the constant term in the velocity-dependent expression $C_d = \frac{c_1}{v} + c_2$. See the note about this in the caption to fig. 6.4. See also the note at the end of sec. 3 on the *accuracy* of the experimental value of c_2 .

In view of our rather accurate results for the sphere and the circular disk, we have good reasons to believe that the results given under fig. 7.1 are rather accurate too - accurate for the models driving with a velocity v between 2 m/s and 10 m/s and with no road surface just underneath. Are they, however, also valid for the full scale cars, driving on a road and with a much higher velocity? Considering the $C_d(v)$ in fig. 6.3 for a sphere, one gets some serious doubts - even if forgetting about the lacking surface of the road (to which we shall return in a subsection below). On the other hand: If the drag for a car behaves as that for the circular disk in fig. 6.4, we *do* have some reason to believe in the extrapolation. The examples of C_d -values collected in fig. 2.1 also supports the belief that our results are reasonable - at least as a first approximation. This belief is further sustained by a look at fig. 7.2 below.

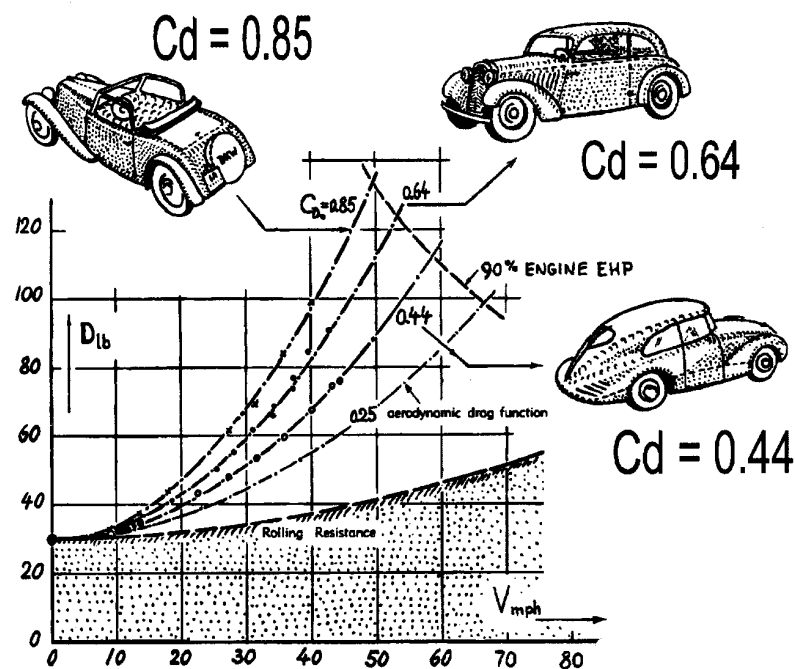


Figure 7.2: Values of C_d for 3 full scale cars resembling the models in fig. 7.1. [5, p. 12 - 7]. They represent three different body designs of the same small 1935 "Auto-Union motorcar: as open roadster, as standard sedan and with a somewhat streamlined body. Since the three cars only differ by their form, they can be assumed to have the same rolling resistance (the lowest graph).

Could we find more evidence that the results obtained for the models are approximately true also for the full scale cars, driving much faster? In sec. 8 we formulate *Reynolds model law*, and see that extrapolation is beyond the limits set by this. We could try to get an answer through a long series of experiments with models for which the full-scale C_d -values are known. I have not yet had the time to do so. What follows here is some intuitive considerations, founded on what I think I have understood from books and talks with persons who know more than I do.

In sec. 5 we saw that the C_d -graph for the circular disk did not show the same sudden drop as that for the sphere. The latter originates from the way the air passes the sphere. The streamlines follow the surface for a while to leave it at a certain point, thereafter forming a more or less turbulent wake. As the velocity increases, this wake becomes more and more narrow and rather suddenly the turbulence fills in a thin layer almost all the way around the sphere. For the circular disk it is different - here turbulence is developed where the streamlines leave the sharp periphery - the border line to the laminar stream has no surface to follow as with the sphere. Hence the constancy of C_d . Elder cars in particular have many protruding parts (licence plates, bumpers, rear view mirrors, its whole underside, ...) and it has irregularities like openings to the wheels. We get

turbulence immediately after these, i.e. a flow pattern that cannot change substantially, along the sides of the car. Such cars tend to take over the C_d -constancy of the disk. The more streamlined modern cars tend to inherit some of the characteristics of the sphere. This gives us some reason to believe that the merry-go-round generally will give reasonable results for bodies with a bad aerodynamic design like the vintage car in fig 7.1. The results for the two other cars indicate that improved streamlining is clearly seen in the measured values of C_d . Note in particular how the open windows in the Ferrari GTO make C_d rise from 0.43 to 0.46. According to Hoerner [5, p. 12-6] open windows in a car like the medium one in fig. 7.2 means an increment in C_d of about 0.03 - 0.04.



Figure 7.3: A very fine model of Lexus LS 400, a luxury car with $C_d \approx 0.30$ in full scale. On the merry-go-round it was found to have $C_d = 0.36$ (closed windows) . With a box for skies on the roof, the value rose to 0.42 . The front area went up from 84.2 to 102.8 cm^2 so that the total air resistance, at the same speed, would go up by the factor $\frac{0.42}{0.36} \cdot \frac{102.8}{84.2} = 1.42$; that is, by 42 % .

Consider finally the car, shown as a model in fig. 7.3. A student of mine got it from an auto dealer together with the following description

The LS400's distinctive styling bespeaks refinement and luxury. Closer inspection reveals the precision of adjoining lines of bonnet and fenders, the exacting alignment of all body panels, window glass and components with their promise of quality and aerodynamic excellence. For today's automotive design relies on aerodynamics to obtain maximum importance from minimum energy input to for good reason. At highway speeds it accounts for half the energy required to propel the car. In order for the total aerodynamic management to be effective, Lexus started from the bottom up. The entire underside was designed for maximum aerodynamic efficiency: A flat engine cover, smooth fairings in front of the tyres and mufflers and airfoil lower suspension arms. Flush mounted body-panels, door handles and light lenses all contribute to a smoother body surface. Each angle of the entire body was scrutinised through the use of computer simulation to determine the optimum shape, which created the minimum of turbulence. The result of this obsessive attention is **an outstanding drag coefficient of 0.29 - 0.30** , low enough to

embarrass the majority of sports cars as well as exceptional coefficients of lift and yawing for greater stability.

The result found by use of the merry-go-round did not agree exactly with the result for the full scale car. However, it was not totally wrong either. Other students of mine have found the value $C_d = 0.36$ for a model of Ferrari Testa-rossa - a value they found confirmed in some magazine. However, I am no longer in possession of all the data from their experiment.

On the influence of the surface of the road

The measurements on model cars described above are all unrealistic in that they move freely in the air without the surface of the road just underneath. Figure 7.4 considers various methods whereby one has dealt with this problem in real wind tunnel test. Readers who wants to know more about this problem are referred to the book by Hoerner [5, p. 12-1] from which the figure is copied.

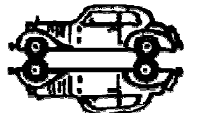
				
Without ground	Fixed plate	Image	Moving belt	Full scale
0.19 89%	0.23 107%	--	0.21 100%	0.24 114%
0.36 97%	0.38 102%	--	0.37 100%	0.38 102%
0.50 97%	0.53 103%	0.53 104%	0.52 100%	0.52 100%
0.52 96%	0.56 103%	0.55 102%	0.54 100%	0.57 104%

Figure 7.4: Four methods whereby one has attempted to make up for the missing road surface in wind tunnel tests on (model-) cars. The full scale values: $C_d = 0.24, 0.38, 0.52$ and 0.57 ; in the last column are obtained either in wind tunnel tests or in road tests¹. The values found on the belt, which moves with same same velocity as the wind, is considered to be rather good in general - although it involves a quite large error for the most streamlined car.

The fixed-plate method could be attempted with the merry-go-round, and so could the third, called *the image method*, in which the symmetry plane between the two models is supposed to imitate the surface of the road.

¹ It is not indicated in Hoernes book which is which. The method of the road test reminds of that of the merry-go-round: The car enters a long vertical piece of road with full speed. The transmission is set in neutral, and the acceleration is determined by reading a clock when the speedometer passes 100, 95, 90, 85, ... *km/h* . Multiplying the cars acceleration with its mass, one gets the air resistance force plus the rolling resistance. The C_d -values given in fig. 7.2 above are actually determined in this way. A set of data for such a determination can also be found in [12]. Here the authors assume that the rolling resistance is a constant - an assumption which according to Hoerner (see the graph for the rolling resistance in fig. 7.2 above) is not quite true. However, driving on an abandoned airport runway, the authors find $C_d = 0.55$ (Toyota Tercel liftback) compared to 0.54 in the literature.